

MANNE LAKSHMI CHAITANYA

AI Engineer · Generative AI · Computer Vision · MLOps
Bangalore, India | +91 8985200204 | mannelchaitanya@gmail.com | [linkedin](#) | [github](#)

PROFESSIONAL SUMMARY

AI Engineer with hands-on experience in Generative AI, Computer Vision and scalable ML systems. Skilled in building real-time face recognition pipelines, RAG-based applications, and deploying production-ready ML APIs using FastAPI, Docker, and AWS. Strong foundation in deep learning, semantic search, and model optimization for edge and cloud environments.

TECHNICAL SKILLS

Languages	Python
ML / Deep Learning	PyTorch, TensorFlow/Keras, Scikit-learn
Computer Vision	OpenCV, CNN, YOLOv8, ML Kit, ArcFace
GenAI / LLM	LangChain, RAG, Vector Databases, LLM Fine-tuning (LoRA/QLoRA), HuggingFace Transformers, Prompt Engineering
Data & Database	NumPy, Pandas, SQL, SQLite
Deployment/MLOps	FastAPI, Flask, Docker, AWS (S3, EC2), MLflow, Git
Soft Skills	Problem-Solving, Team Collaboration, Effective Communication, Time Management

EXPERIENCE

- AI Engineer Intern · Techsoc Technologies Pvt. Ltd. · Bangalore

Jan 2026 – Present

 - Built a real-time face recognition attendance application using ML Kit (face detection) and ArcFace (512-d embeddings) with cosine similarity and optimized threshold tuning for accurate identity matching on live video streams; reduced latency by 30%
 - Designed, developed, and maintained backend services and REST APIs for attendance management, handling user data, embeddings, and daily attendance logs with cloud-based deployment on AWS (S3, EC2)
 - Developed multi-class object detection pipelines with YOLOv8 and OpenCV across image and video data sources
 - Converted and optimized trained PyTorch models to TFLite format for deployment on Android edge devices, reducing on-device inference latency
- AI Consultant Intern · Rubixe AI Solutions · Bangalore

May 2025 – Dec 2025

 - Designed, trained, and evaluated ML models (classification, regression) using Python, NumPy, Pandas, and Scikit-learn on real-world datasets
 - Implemented CNN-based architectures for image classification and object detection; tuned performance via hyperparameter optimization and cross-validation
 - Built NLP pipelines for text preprocessing, sentiment analysis, and sequence classification using Keras and HuggingFace Transformers

PROJECTS

- AI Document Assistant — Generative AI & RAG — 82% accuracy on response quality evaluation

 - Architected a full RAG pipeline using LangChain and vector databases for semantic retrieval over multi-document inputs, enabling context-aware LLM responses
 - Implemented embedding-based search with FAISS and optimized chunking strategies to improve retrieval relevance and reduce hallucinations
 - Developed FastAPI-based APIs for document ingestion and query handling; designed scalable pipeline supporting multiple documents and efficient semantic search
- Face Recognition System — Computer Vision — Processed real-time video at 25 FPS

 - Developed an end-to-end face recognition pipeline using ArcFace embeddings with cosine similarity for accurate face identification and verification
 - Built and optimized real-time video inference using OpenCV with scalable embedding storage (SQLite), threshold-based matching, confidence scoring, and unknown-person detection
- Diabetes Prediction System — End-to-End MLOps — 84% classification accuracy on held-out test data

 - Trained and benchmarked Logistic Regression, Random Forest, and XGBoost classifiers; optimized with GridSearchCV and handled class imbalance via SMOTE
 - Deployed model as a FastAPI REST API containerized with Docker on AWS EC2; tracked experiments with MLflow and versioned artifacts on AWS S3

EDUCATION

Bachelor of Engineering — Computer Science & Engineering
Bapatla Engineering College, Bapatla

CGPA: 6.15 · 2021 – 2025

CERTIFICATIONS

Data Science & AI Consultant Certification · Rubixe AI Solutions, 2025